Special Relativity
Module 1: Introduction

Modern physics, common sense, and the place of special relativity

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Special relativity revises the classical description of space, time, and motion while explaining the success of Newtonian mechanics at low speeds. This introduction develops its physical motivation, states its guiding principles, and examines its place in modern physics. Worked examples connect the basic ideas with atmospheric muons, particle accelerators, and satellite navigation.

By the end of the module, you should be able to explain why everyday intuition has a limited domain of validity, identify the conflict between Galilean kinematics and Maxwell's electrodynamics, state Einstein's two postulates, and place special relativity in relation to Newtonian mechanics, quantum theory, and general relativity. You should also be able to explain the role of relativity in atmospheric muons, accelerators, and satellite navigation.

We like to point that, in this Module, the student may not completely understand all the presented mathematical formulas. This is not important, since this is a prelude to what will come. All presented formulas will derived later, in due time.

1 The cosmology of the senses

Our immediate experience covers a narrow range of lengths, speeds, and times. We handle objects comparable to our own size, watch bodies moving much more slowly than light, and ordinarily compare clocks whose relativistic rate differences are too small to notice. It is unsurprising that the concepts developed from such experience work well within this range.

Trouble arises when these concepts are promoted to universal truths. Two events that seem simultaneous to one observer need not be simultaneous to another. A clock's elapsed time depends on its path through spacetime. Neither statement is a contradiction: each challenges a familiar assumption about how measurements made by different observers must be related.

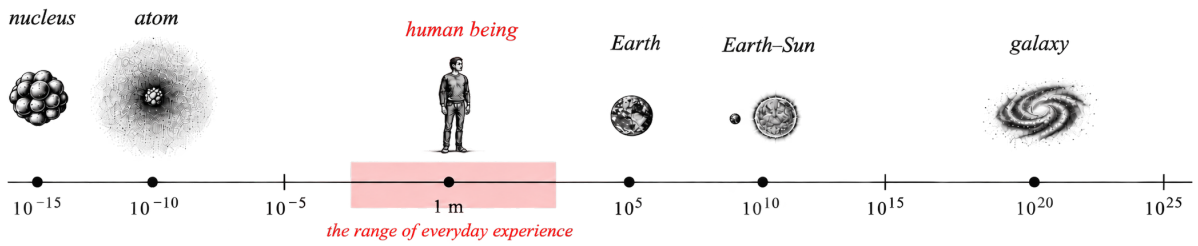


Figure 1. A schematic overview of physical length scales. The object drawings and their placement are schematic. For quantitative orientation, Earth's radius is about 6.4×10^6 m, the mean Earth–Sun distance is about 1.5×10^{11} m, and a large galaxy spans roughly 10^{21} m. The highlighted region represents the limited range of ordinary human experience.

Modern physics does not ask us to abandon reasoning. It asks us to test the assumptions on which reasoning is based. A physical theory must agree with observations, including observations made

outside the circumstances that shaped our intuitions. When a new theory reproduces a familiar theory in a well-defined limit, it also explains why that familiar theory worked.

2 Newtonian mechanics and inertial frames

Newtonian mechanics gave mathematical form to a successful description of motion: positions change in a three-dimensional Euclidean space, all observers share a common time, and forces determine accelerations. Its scope extends far beyond daily experience, from machinery to planetary motion. Nevertheless, its assumptions about time and motion are not exact properties of nature.

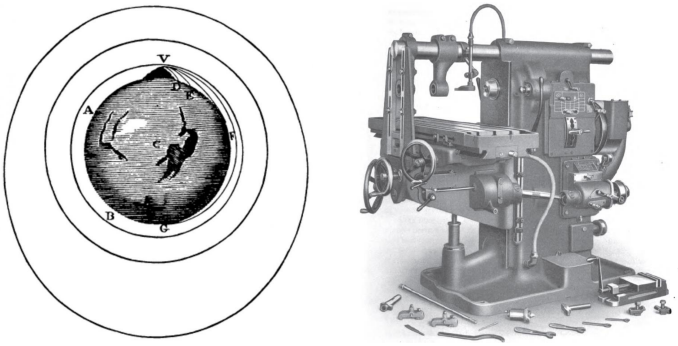


Figure 2. Newtonian celestial mechanics and its technological reach. Machine image: Brown & Sharpe Mfg. Co., *Practical Treatise on Milling and Milling Machines.*

An inertial frame is a reference frame in which a freely moving test body has constant velocity. Ideally, it is realized by a system of spatial coordinates and synchronized clocks, with negligible external disturbances. No actual laboratory is perfectly inertial, but the approximation can be excellent over a specified region and duration.

Suppose a frame S' moves with constant velocity v along the x axis of S. Align the axes and let the origins coincide at $t = t' = 0$. The Galilean transformation is

$$x' = x - vt, \qquad y' = y, \qquad z' = z, \qquad t' = t. \tag{1}$$

Differentiation gives $u'_x = u_x - v$ and $a'_x = a_x$. Thus, for forces that transform appropriately, Newton's equation $\mathbf{F} = m\mathbf{a}$ has the same form in both frames. The equality $t' = t$ encodes absolute time.

The principle of relativity in mechanics says that an isolated mechanical experiment cannot determine the uniform velocity of its entire laboratory. A pendulum and a spring–mass system behave in the same way in a laboratory at rest and in one translating uniformly, provided their physical conditions are the same. An experiment can detect acceleration or rotation, and it can measure velocity relative to an external object. The principle concerns the absence of an internally measurable state of absolute uniform motion.

3 The conflict between mechanics and electrodynamics

In vacuum, Maxwell's equations imply electromagnetic waves satisfying

$$\nabla^2 \mathbf{E} - \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} = 0, \qquad c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}. \tag{2}$$

Here ε_0 and μ_0 are the vacuum electromagnetic constants in SI units. The identification of this wave speed with the measured speed of light connected optics with electricity and magnetism.

Now consider a light pulse traveling along $+x$, so that $x = ct$. Under Eq. (1), its trajectory would be

$$x' = (c - v)t', \qquad u'_x = c - v. \tag{3}$$

This differs from the speed c predicted by the same vacuum Maxwell equations in S'. Consequently, unmodified Maxwell electrodynamics, Galilean transformations, and equivalence of all inertial frames cannot all be retained.

The nineteenth-century aether hypothesis treated light as a disturbance of a medium with a preferred rest frame. That proposal offered a possible answer to the question of which frame should measure the Maxwell wave speed. Experiments searching for motion relative to such a medium, including the Michelson–Morley experiment of 1887, constrained this picture. The path to special relativity involved both this experimental background and the conceptual structure of electrodynamics; it was not the consequence of one isolated measurement.

3.1 The magnet and conductor

Einstein opened his 1905 paper with the relative motion of a magnet and a conductor [6]. In the conductor's rest frame, motion of the magnet produces a time-dependent magnetic field and an electric field obeying Faraday's law,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{4}$$

In an idealized magnet rest frame, the magnetic field is static and the conductor moves. The magnetic contribution to the force on its charges is $q\mathbf{u} \times \mathbf{B}$. The complete force law in either frame is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}). \tag{5}$$

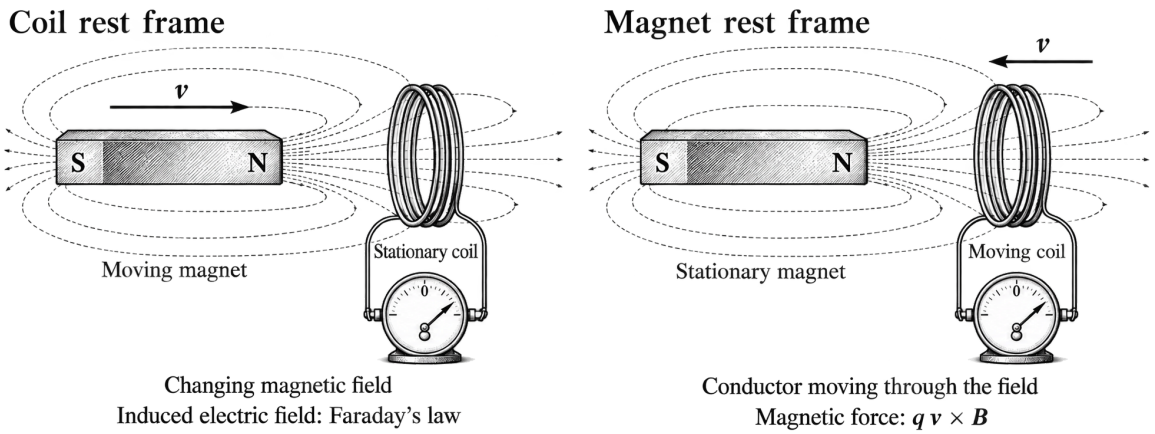


Figure 3. The same induction experiment described in two inertial frames. The displayed velocity refers to the moving body in each panel. The electric and magnetic contributions are frame dependent; both descriptions use Maxwell's equations and the Lorentz force.

For a circuit whose elements move with local velocity $\mathbf{u}$, the electromotive force can be written

$$\mathcal{E} = \oint_{C(t)} (\mathbf{E} + \mathbf{u} \times \mathbf{B}) \cdot \mathrm{d}\boldsymbol{\ell}. \tag{6}$$

For a stationary circuit, $\mathbf{u} = 0$ and the electric term supplies the emf. In the moving-conductor description, the magnetic term can supply it. The usual magnetic-flux rule follows for suitable circuits from this same expression and Maxwell's equations.

The apparent asymmetry was therefore in the explanation given to the two situations. Relativity makes the division into electric and magnetic effects dependent on the observer. It does not introduce one set of Maxwell laws for a moving magnet and another for a moving conductor. Nor must forces or currents have identical numerical values in every frame: the complete descriptions must be related consistently by the transformation laws.

4 Einstein's postulates and the new kinematics

Einstein's formulation rests on two physical postulates:

1. The laws of physics have the same form in every inertial frame.
2. Light in vacuum propagates with the same speed c in every inertial frame, independently of the motion of its source.

The second statement concerns locally measured vacuum light speed. It does not assert that light has speed c in material media or that every coordinate speed in an accelerated coordinate system equals c.

With standard Einstein synchronization, the transformation between the two frames introduced in Eq. (1) becomes

$$x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{vx}{c^2}\right), \qquad y' = y, \quad z' = z, \tag{7}$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{8}$$

We will derive these equations later. Their immediate conceptual lesson is that the transformation of time involves position. If two spatially separated events are simultaneous in S, then

$$\Delta t = 0 \quad \Longrightarrow \quad \Delta t' = -\gamma \frac{v\Delta x}{c^2}. \tag{9}$$

They are generally not simultaneous in S'. This is a statement about assigned event times after synchronization, not merely about the different times at which light from the events reaches an observer's eyes.

Time dilation, length contraction, and the relativistic velocity law follow from the same transformation. The energy–momentum relation is part of the corresponding dynamics. In particular, $E_0 = mc^2$ is the energy of a body in its rest frame; for a moving particle, its total energy is $E = \gamma mc^2$.

4.1 A first view of spacetime

Minkowski's geometric formulation combines the coordinates into spacetime. For two events in inertial coordinates, define

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \tag{10}$$

Lorentz transformations preserve this interval. Although observers disagree about its separate temporal and spatial parts, they agree on its value.

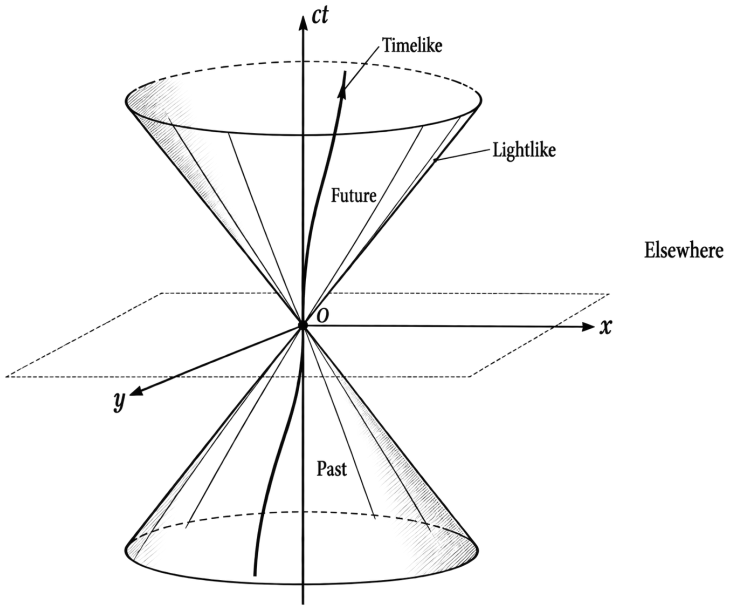


Figure 4. The light cone of an event O. One spatial dimension is suppressed. Light emitted at O lies on the future cone; light arriving at O lies on the past cone. The region labeled "elsewhere" is spacelike separated from O.

For a timelike separation, $\Delta s^2 > 0$, a massive body can connect the events. A lightlike separation has $\Delta s^2 = 0$; a spacelike separation has $\Delta s^2 < 0$. The cones therefore express causal structure, rather than the visible boundary of an expanding material object. The proper time along a timelike path is the time registered by a clock traveling along it.

5 A principle that constrains other laws

The relativity principle is broader than a law for a particular force or particle. It requires candidate physical laws to remain consistent when expressed in different inertial frames. In this sense it is a condition on the form of laws, including laws not yet discovered.

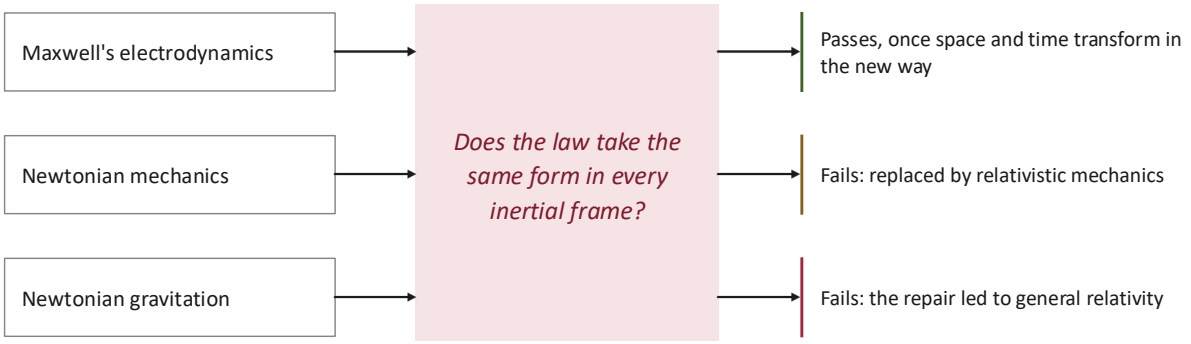


Figure 5. The relativity principle as a test of physical laws. The verdict for Newtonian mechanics depends on which spacetime transformation is being used: it is Galilean invariant, but it is not the exact mechanics of Lorentzian spacetime.

Maxwell's equations satisfy this requirement under Lorentz transformations, with electric and magnetic fields transforming together. Newtonian mechanics must be replaced by dynamics with relativistic momentum $\mathbf{p} = \gamma m \mathbf{u}$. Its low-speed success is retained as an approximation. Newton's instantaneous gravitational interaction is also incompatible with a fundamental Lorentz-invariant

description of causal propagation. General relativity supplies a dynamical account of gravity and recovers Newtonian gravitation in an appropriate weak-field, slow-motion limit.

Invariance of a law does not mean invariance of every quantity in it. Velocity, energy, electric field, and coordinate time can differ between observers. The equations relating them must transform consistently, and measurable outcomes must agree when the observers compare descriptions of the same events.

6 Historical change and public reception

The transition from Newtonian physics to modern physics occurred through several developments with different motivations. Maxwell's synthesis linked light and electromagnetism; Planck's 1900 work addressed thermal radiation; Einstein's 1905 work revised inertial kinematics; Minkowski emphasized its geometry in 1908; Einstein completed general relativity in 1915; and the development of quantum mechanics in 1925–1927 reorganized the description of atoms and measurements.

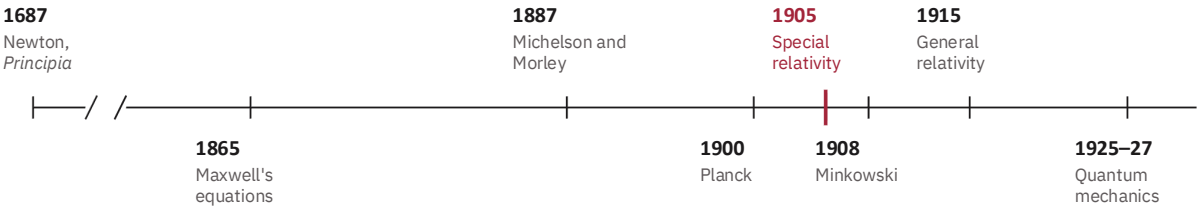


Figure 6. Selected milestones in the development of modern physics. Dates identify selected milestones, not instantaneous changes in scientific consensus.

Quantum mechanics challenged classical assumptions about physical properties and measurement. The stronger statement that it abolished objective reality would depend on an interpretation of quantum theory. For our purpose, the relevant point is that successful theories can require concepts quite different from those suggested by ordinary experience.

Relativity also became part of public culture. The cartoon and film image in Fig. 7 document this reception. Herman Hupfeld's song *As Time Goes By* (1931) [11], later associated with *Casablanca* (1942), refers to Einstein in its introductory verse. Such references show how widely the name of the theory circulated, rather than demonstrating public understanding of its content.

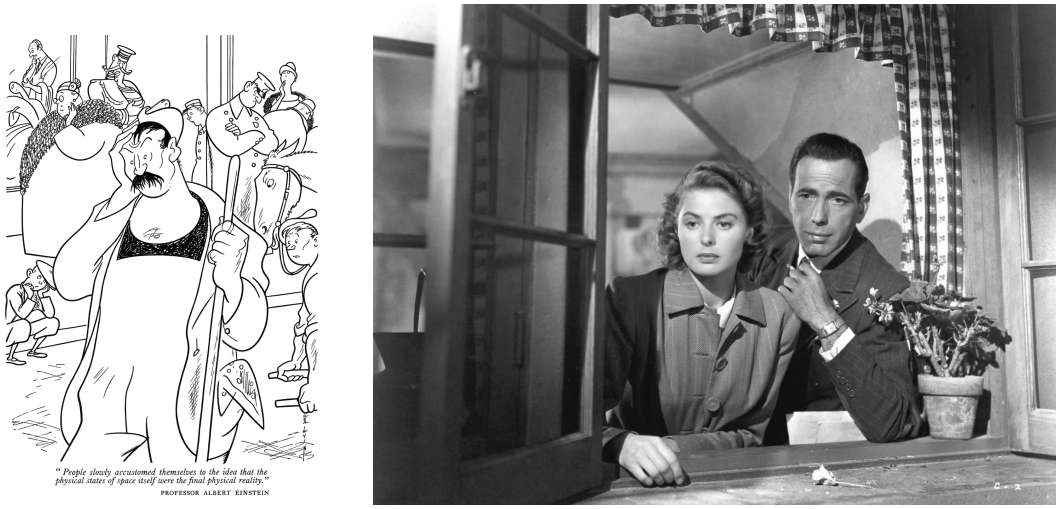


Figure 7. Relativity in public culture. Left: Rea Irvin, *The New Yorker*, 16 March 1929. Right: a scene from *Casablanca* (1942); image source: Wikimedia Commons.

7 How theories contain earlier descriptions

A later theory need not make an earlier one useless. It can explain why the earlier theory succeeds in a restricted domain. Special relativity recovers Newtonian mechanics when v/c is sufficiently small. Quantum theory can recover classical behavior when relevant actions are large compared with $\hbar$ and the state and experimental conditions support a classical approximation. Quantum field theory combines quantum principles with special relativity and allows particle creation and annihilation.

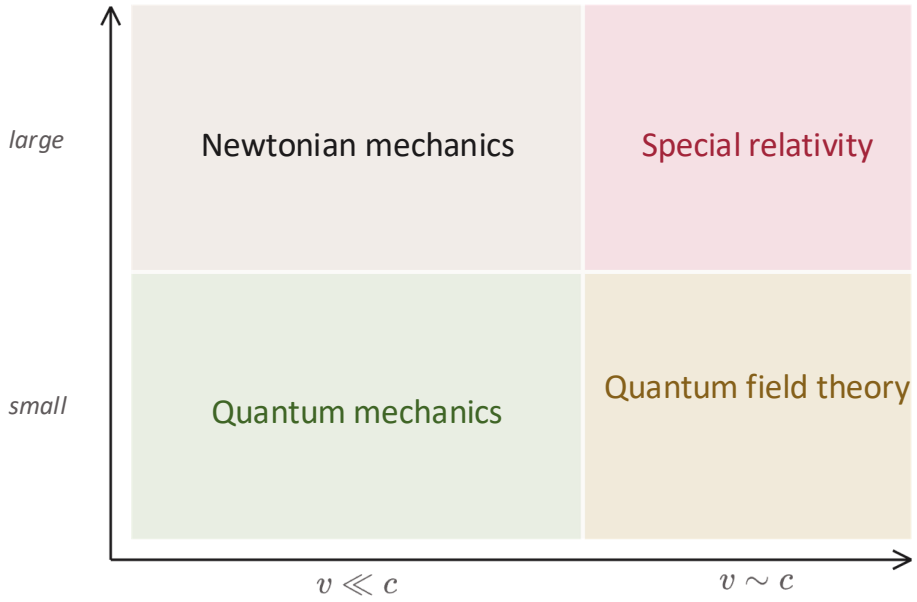


Figure 8. Domains of physical theories. This is a schematic map: "small" and "large" do not refer only to geometric size. Quantum effects depend on the relevant action scale, and relativistic effects on energy and velocity scales.

For a massive particle, expansion of Eq. (8) gives

$$\gamma = 1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4} + \cdots . \tag{11}$$

Its kinetic energy is therefore

$$K = (\gamma - 1)mc^2 = \frac{1}{2}mv^2 + \frac{3}{8}\frac{mv^4}{c^2} + \cdots . \tag{12}$$

The leading term is the familiar Newtonian result. The approximation has a quantifiable error rather than an unrestricted range of validity.

The outstanding problem of quantum gravity is not simply to join a theory of "large things" to a theory of "small things." General relativity describes classical dynamical spacetime; quantum theory describes matter and fields. Both may be needed, for example, where quantum fluctuations of the gravitational field become important. Quantum fields on a classical curved background provide a useful partial framework, but are not a complete quantum theory of spacetime. No experimentally established complete theory currently settles this problem.

8 Why relativistic corrections were hard to notice

The importance of a correction depends both on the dimensionless parameter $\beta = v/c$ and on measurement precision. For $\beta \ll 1$, $\gamma - 1 \simeq \beta^2/2$. At Earth's orbital speed, approximately $30\,\mathrm{km\,s^{-1}}$, $\beta \simeq 10^{-4}$ and $\gamma - 1 \simeq 5 \times 10^{-9}$. Such corrections are negligible for many mechanical calculations, yet can matter for precision clocks.

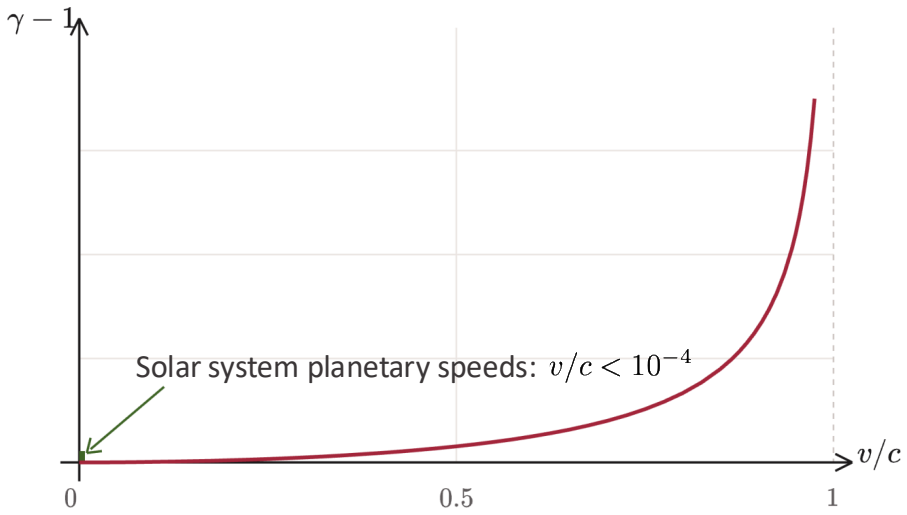


Figure 9. The relativistic correction $\gamma - 1$ as a function of v/c. The rapid increase near $v = c$ contrasts with the small correction at planetary speeds.

For comparison, at $v = 0.8c$ one obtains $\gamma = 5/3$. A freely moving clock that accumulates a proper time of 1 s between two events then has $5/3$ s between those events in the inertial frame where its speed is $0.8c$. The time-dilation relation is

$$\Delta t = \gamma \Delta \tau. \tag{13}$$

The two events lie on the moving clock's worldline. This specification prevents the ambiguous statement that one observer's "time itself" is universally slower than another's.

Theoretical reasoning developed within the experimental setting of 1905. Electromagnetic and optical evidence already constrained possible theories, and fast-electron measurements were under discussion. Einstein's achievement was to reorganize the description around physical principles and operational definitions, rather than merely fit one new anomaly.

9 Where special relativity stands

Special relativity is the spacetime framework for nongravitational relativistic physics in flat spacetime. It is built into relativistic quantum field theory and the standard description of particle physics. Its low-speed limit contains Newtonian mechanics, while general relativity recovers its local structure in sufficiently small freely falling laboratories.

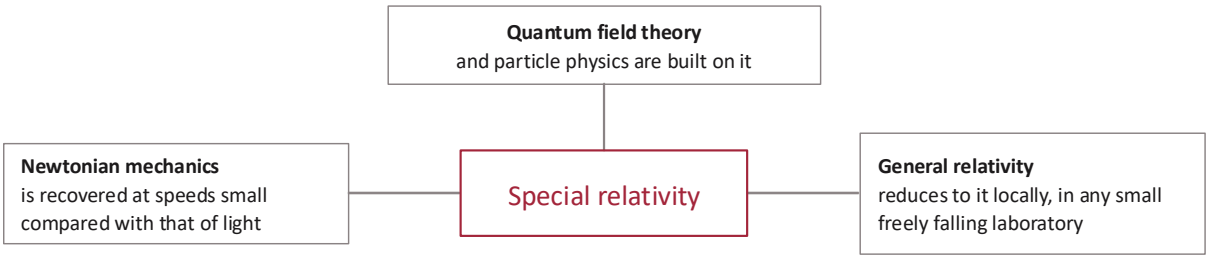


Figure 10. Special relativity and its relationships with other theories. The arrows represent limiting relationships and incorporation into broader theories, not a sequence in which special relativity is discarded.

The word "local" matters. In a small freely falling laboratory, tidal effects can be made sufficiently small and nongravitational experiments are described approximately by special relativity. Over an extended curved region, there need not be a single global inertial frame. Accelerated observers can also be studied within special relativity: acceleration alone does not imply spacetime curvature.

10 Experimental example: atmospheric muons

Cosmic rays interact with atmospheric nuclei and generate secondary particles, including pions whose decays produce muons. A muon at rest has a mean lifetime of approximately $\tau_0 = 2.2\,\mu\text{s}$. At nearly the speed of light, the distance $c\tau_0$ is only about 660 m. Many muons are produced several kilometres above the ground, so their survival provides a useful application of relativistic clock comparisons.

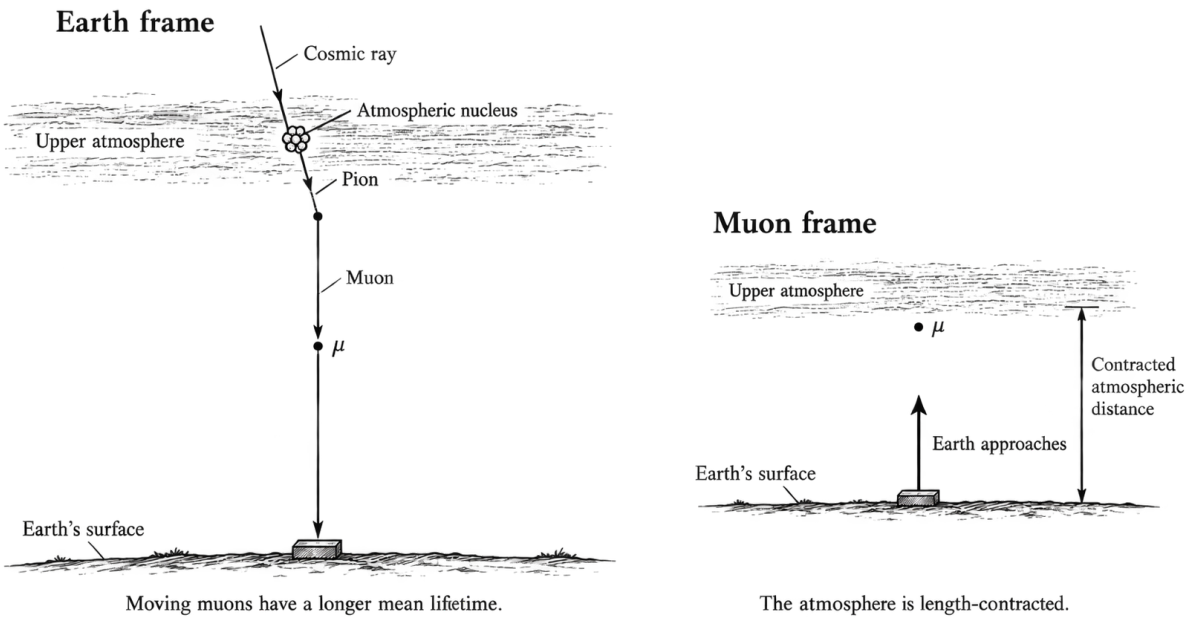


Figure 11. Atmospheric muons described in two inertial frames. In the Earth frame, moving muons have a longer mean lifetime. In an individual muon's rest frame, the distance through the atmosphere is shorter. These are consistent descriptions of the same production and detection events.

As an illustrative model, take a vertical path $L = 10\,\text{km}$ and a constant muon speed $v = 0.998c$.

Neglect energy loss and scattering. Then

$$\gamma \simeq 15.82, \qquad \Delta t = \frac{L}{v} \simeq 33.42\,\mu\text{s}, \qquad \Delta\tau = \frac{L}{\gamma v} \simeq 2.11\,\mu\text{s}. \tag{14}$$

The decay law gives the survival probability

$$P = \exp\left(-\frac{\Delta\tau}{\tau_0}\right) = \exp\left(-\frac{L}{\gamma v \tau_0}\right) \simeq 0.38. \tag{15}$$

Omitting time dilation would instead give roughly 2.5×10^{-7} for these same assumed height and speed. In the muon frame, the initial Earth-frame distance is contracted to $L' = L/\gamma \simeq 632\,\text{m}$, and the Earth reaches the muon after $L'/v \simeq 2.11\,\mu\text{s}$.

The muon discovery, conventionally dated to Anderson and Neddermeyer in 1936 with publication in 1937 [7], should be distinguished from later decay experiments testing relativity. Rossi and Hall's 1941 work is an important early time-dilation test [8]. Actual atmospheric flux calculations also account for energy distributions, production altitudes, and interactions in the atmosphere.

11 Experimental example: particle accelerators

Accelerators provide controlled beams for which relativistic dynamics is essential. For a particle of rest mass m and speed v,

$$E = \gamma mc^2, \qquad p = \gamma mv, \qquad E^2 = p^2c^2 + m^2c^4. \tag{16}$$

As v approaches c, added energy produces an increasingly large increase in momentum and a diminishing increase in speed. For $K \gg mc^2$, $E \simeq pc$, while v remains below c for a massive particle.

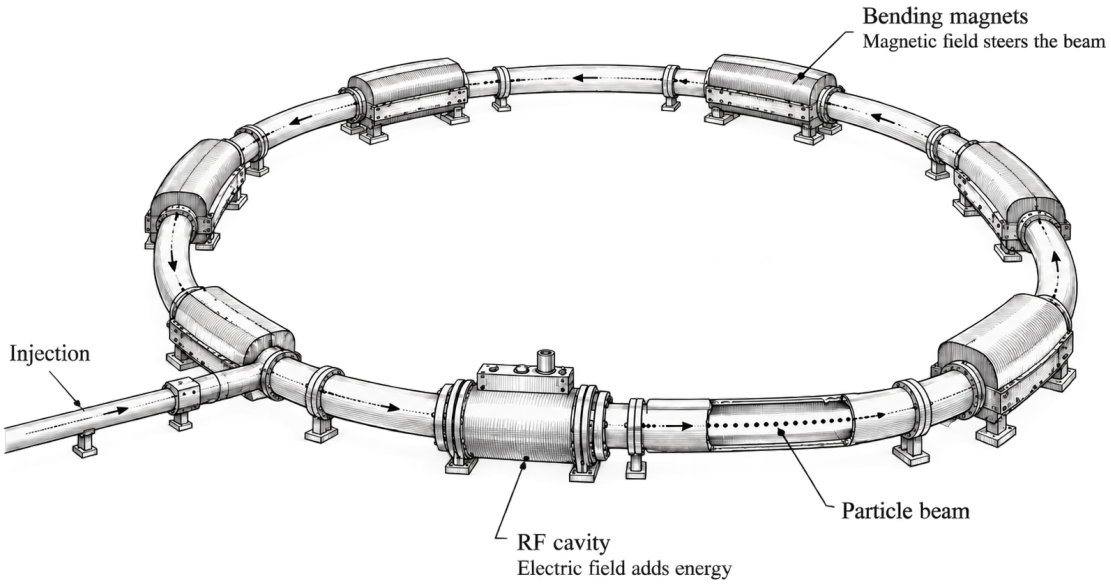


Figure 12. A schematic ring accelerator. Radio-frequency cavities transfer energy to the beam and bending magnets guide it. This is a generic RF ring illustration, not a drawing of Kerst's 1940 betatron, whose accelerating electric field was induced magnetically.

For circular motion perpendicular to a uniform magnetic field, magnetic bending relates momentum to radius through

$$p = |q|Br. \tag{17}$$

Using $p = mv$ at relativistic energies would give the wrong relationship between beam energy, field, and bending radius. The corresponding angular frequency is $\omega = |q|B/(\gamma m)$, which also explains why relativistic effects complicate the fixed-frequency operation of a conventional cyclotron.

Kerst's first successful betatron operated in 1940 [9]. For electrons with $K = 2.3\,\mathrm{MeV}$ and rest energy $m_e c^2 \simeq 0.511\,\mathrm{MeV}$,

$$\gamma = 1 + \frac{K}{m_e c^2} \simeq 5.50, \qquad \frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}} \simeq 0.983. \tag{18}$$

Modern machines explore much larger ratios of beam energy to particle rest energy.

12 Experimental example: satellite navigation

GPS receivers infer position and time using signals from satellites with accurate clocks. A timing error δt corresponds to a signal-range error of order $c\delta t$. Consequently, a clock-rate difference too small to notice in ordinary experience can be significant for navigation.

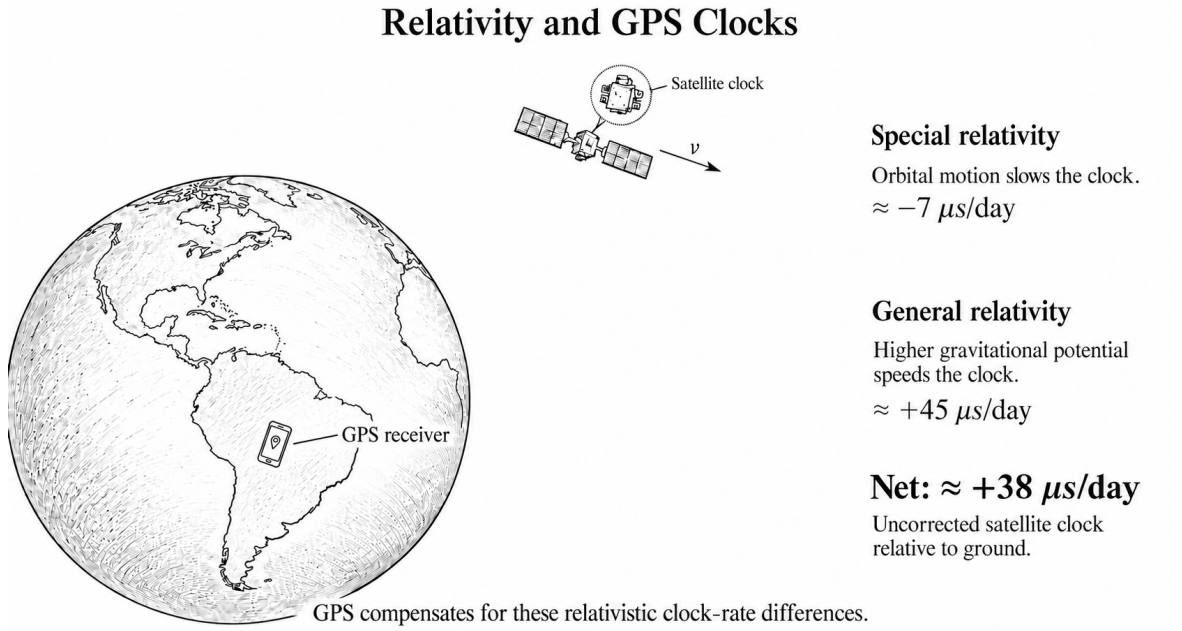


Figure 13. Relativistic clock-rate effects in GPS. The quoted daily clock-rate shifts are approximate values for nominal GPS orbits relative to clocks on Earth's geoid. The positive net shift is the uncorrected tendency of the satellite clocks to gain time.

Two leading effects have opposite signs [10]. Orbital motion slows a satellite clock by about $7\,\mu\mathrm{s}$ per day relative to the reference comparison. The higher gravitational potential at satellite altitude makes it run faster by about $45\,\mu\mathrm{s}$ per day. Together they give approximately $+38\,\mu\mathrm{s}$ per day.

In a weak, stationary gravitational field and for speeds much smaller than c, the clock rate has the approximation

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} \simeq 1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2}, \tag{19}$$

where Φ is the Newtonian gravitational potential and t is the chosen coordinate time. Comparing two clocks requires their potential and velocity differences in a consistent reference system. The gravitational term belongs to general relativity; the kinematic term is the special-relativistic effect.

A daily uncompensated offset of 38 μs corresponds to a range scale of about 11.4 km. This is an indication of the importance of timing accuracy, not an exact prediction of a receiver's position error, which also depends on satellite geometry and clock estimation. GPS incorporates relativistic frequency adjustments and propagation corrections, including effects associated with Earth's rotation. NAVSTAR 1, launched in 1978, marks the first Block I GPS satellite; the navigation system developed through later generations.

13 A route through special relativity

The course begins with classical kinematics because its assumptions make the contrast with relativistic kinematics precise. We will define measurements, introduce Einstein synchronization, derive transformations, and interpret their consequences geometrically. Later modules develop four-vectors, relativistic mechanics, accelerated observers, and electromagnetism.



Figure 14. A route from classical to relativistic kinematics, Einstein's postulates, and Minkowski spacetime.

An additional foundations unit examines transformations without assuming the light-speed postulate at the outset. Homogeneity, isotropy, relativity, and appropriate continuity and composition assumptions constrain possible kinematics. They do not by themselves identify a finite invariant speed with the measured speed of light; that step requires physical input.

For each calculation, specify the frame, the events being compared, and the measurement procedure. Distinguish a clock's proper time from coordinate time, a simultaneous length measurement from visual appearance, and a frame-dependent quantity from an invariant. Spacetime diagrams become useful only when their axes, worldlines, and selected events are identified.

13.1 Questions for review

1. Why does the relativity principle prohibit measuring absolute uniform motion using an isolated experiment, while allowing detection of acceleration?
2. Use the Galilean transformation to find the speed assigned to a pulse with trajectory $x = ct$. Which assumptions create the conflict with Maxwell's electrodynamics?
3. In the magnet–conductor experiment, what changes between frames: Maxwell's equations, the field components, or both?
4. Two events have $\Delta t = 0$ and $\Delta x \neq 0$ in S. What does Eq. (7) predict for their time separation in S'?
5. Estimate $\gamma - 1$ at $v = 10^{-3}c$. Must such a correction be irrelevant to every experiment?
6. Explain the atmospheric-muon example in both frames without claiming that the muon's own decay mechanism changes in its rest frame.
7. Find γ and v/c for an electron with kinetic energy $4m_ec^2$. Compare its speed with the Newtonian prediction.
8. Why does a full account of GPS require general relativity as well as special relativity?

13.2 Brief answers and guidance

1. Inertial frames are related by uniform-velocity transformations under which the physical laws have the same form. Acceleration can be detected through local accelerometer readings; rotation can also produce internal effects.
2. $x' = (c - v)t'$, hence $u'_x = c - v$. Universal validity of the vacuum Maxwell equations conflicts with this result if Galilean transformations are retained.
3. The components of $\mathbf{E}$, $\mathbf{B}$, and charge velocities change. Maxwell's equations and the Lorentz force retain their form.
4. $\Delta t' = -\gamma v \Delta x / c^2$. Spatially separated simultaneity is generally frame dependent.
5. $\gamma - 1 \simeq 5 \times 10^{-7}$. Whether this is negligible depends on the observable, elapsed time, and experimental precision.
6. The Earth frame assigns the moving muon a dilated mean lifetime; the muon frame assigns a contracted atmospheric distance. Both yield the same elapsed proper time and survival probability.
7. $\gamma = 5$ and $v/c = \sqrt{24/25} \simeq 0.9798$. The Newtonian formula would give $v/c = \sqrt{8}$, showing its failure at this energy.
8. Motion changes clock rates, and different gravitational potentials also change clock rates. Signal propagation and the rotating Earth require additional consistent timing corrections.

Further reading and references

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